

KVM

KVM (Kernel-based Virtual Machine) is the preferred hypervisor of infrastructure solutions like OpenStack or openQRM, and enjoys a good reputation within the open source community. It is a full virtualisation solution for Linux on x86 hardware containing virtualisation extensions. It consists of a loadable kernel module, *kvm.ko*, which provides the core virtualisation infrastructure and a processor-specific module, *kvm-intel.ko* or *kvm-amd.ko*. Using KVM, one can use multiple virtual machines running unmodified Linux or Windows images. Each virtual machine has private virtualised hardware - a network card, disk, graphics adapter, etc.

Docker

Docker is an open platform for building, shipping and running distributed applications. It gives programmers, development teams and operations engineers the common toolbox they need to take advantage of the distributed and networked nature of modern applications. The container technology, which was created as a by-product during the development of the dotCloud Platform-as-a-Service, is currently experiencing a strong momentum and gets support from large players like Google, Amazon Web Services and Microsoft. Docker enables the loosely coupled movement of applications that are bundled in containers, across several Linux servers, thus improving application portability. At first glance, Docker looks like a pure tool for developers. From the point of view of an IT decision-maker, however, it is definitely a strategic tool for optimising modern application deployments. Docker helps to ensure the portability of an application, to increase the availability and to decrease the overall risk.

Apache Mesos

Mesos rose to be a top-level project of the Apache Software Foundation. The Mesos kernel runs on every machine and provides applications (e.g., Hadoop, Spark, Kafka, Elastic Search, etc) with APIs for resource management and scheduling across entire data centre and cloud environments. It was conceived at the University of California in Berkeley and helps to run applications in isolation from one another. At the same time, the applications are dynamically distributed on several nodes within a cluster. Mesos can be used with OpenStack and Docker. Popular users are Twitter and Airbnb.

Deltacloud

Deltacloud is an open source project started last year by Red Hat. It is now an Apache incubator project. Deltacloud abstracts the differences between clouds and maps a cloud client's application programming interface (API) into the API of a number of popular clouds. As a result, Deltacloud is a way of enabling and managing a heterogeneous cloud

virtualisation infrastructure. It allows for any certified virtualised environment to be managed from one common management interface. And it does this by enabling different virtual machines to be transferred or migrated in real-time from one virtualisation capacity to another. If an enterprise is already using IBM Tivoli or HP OpenView, Deltacloud can be integrated.

OpenNebula

OpenNebula is an open source toolkit for cloud computing. It allows you to build and manage private clouds with Xen, KVM and VMware ESX, as well as hybrid clouds with Amazon EC2 and other providers through Deltacloud adaptors. The remote public cloud provider could be a commercial cloud service provider such as Amazon, or it could be a partner private cloud running a different OpenNebula instance. 

References

- [1] https://en.wikipedia.org/wiki/Cloud_computing_security
- [2] <http://searchcloudcomputing.techtarget.com>

By: Anjali Menon

The author is an open source enthusiast and a Linux kernel contributor. You can contact her at cse.anjalimenon@gmail.com.

CORE EMBEDDED SOFTWARE DEVELOPMENT COURSES

School of Embedded Software

www.schoolofembedded.org

COURSES:

- Embedded System Programming on Arm Cortex-M platform
- Arm Cortex-M - Architecture, Internals and Programming
- Embedded OS - Architecture, Internals and Programming
- Real-Time OS - Architecture, Internals and Programming
- Embedded OS - Board Support and Drivers Development
- Linux Internals for Driver Development
- Embedded Linux/ Kernel Internals
- BeagleBone/Black hw platforms - New Release
- Embedded Linux/ Device Drivers
- BeagleBone/Black hw platforms - New Release
- Embedded Linux/Real-time
- BeagleBone/Black hw platforms - New Release

VENUE:

School of Embedded Software Development
Kalyan Nagar, HRBR Layout, Bangalore - 560043

Faculty : **Babu Krishnamurthy**

Email : babu_krishnamurthy@schoolofembedded.org

Phone : +91-98453-65845/080-41207855

COURSE DETAILS:

http://www.schoolofembedded.org/course_s
<http://www.schoolofembedded.org/events-default>